Melika Shahghadami

Technical Artist | 3D Artist | Unity | VR | Unreal Engine

Technical Art • 3D Modeling • Unity & Unreal Engine • VR • Optimization

Unity & Unreal Engine | Technical Art & Optimization | VR • Meta Quest 3



Skills

3D Art

High/Low Poly Modeling • Retopology • UV Mapping • PBR Texturing • Baking • Materials • Mesh Optimization

Technical Art

Unity URP • Shader Graph • VFX / Particle Systems • Scene & Lighting Setup • Profiling & Performance Optimization • Editor Tools

Character Art

Rigging • Skinning • Animation Controllers

VR

Meta Quest 3 Optimization • Real-time Performance

Software

Primary

Blender | Unity | Unreal Engine | Substance Painter

Additional

Photoshop | Maya | Character Creator | iClone | PureRef | Illustrator

Key Projects

- **Industrial Warehouse — Unity URP**
  PBR Materials • Volumetric Lighting • Environment Art
- **VR Driving Simulator — Unity / Meta Quest 3**
  Day/Night Cycle • Environment Art • Optimization
- **Multiplayer Shooter — Unity / VR / FPS**
  Level Design • Cover Placement • Sightlines
- **Cinematic Environment — Blender / Cycles**
  PBR • Volumetric Fog • Particle Systems

About Me

Technical Artist and 3D Artist with 3+ years of experience producing and preparing real-time assets for games, simulators and VR projects. Experienced in modeling and optimization, PBR texturing, rigging, and technical asset implementation in Unity and Unreal Engine. Focused on balancing visual quality and performance, especially in VR projects and on Meta Quest 3.

Work Experience

Ario Sorun — 3D Designer & Technical Artist — 2023 – Present

- Asset implementation in Unity, scene layout, material setup and lighting.
- Mesh, texture and draw call optimization to improve performance in VR projects on Meta Quest 3.
- Developing and standardizing the asset production pipeline for better coordination between Art and Development teams.

Pishgaman Lotus — 3D Designer & Godot Programmer — 2022 – 2023

- Environment and modular 3D asset design for game projects.
- Collaborated with the development team on technical implementation and final build preparation.

Tarsim Dadeh — Java Developer — 2021 – 2022

- Developed and maintained software modules in Java; debugging and code optimization.

Education

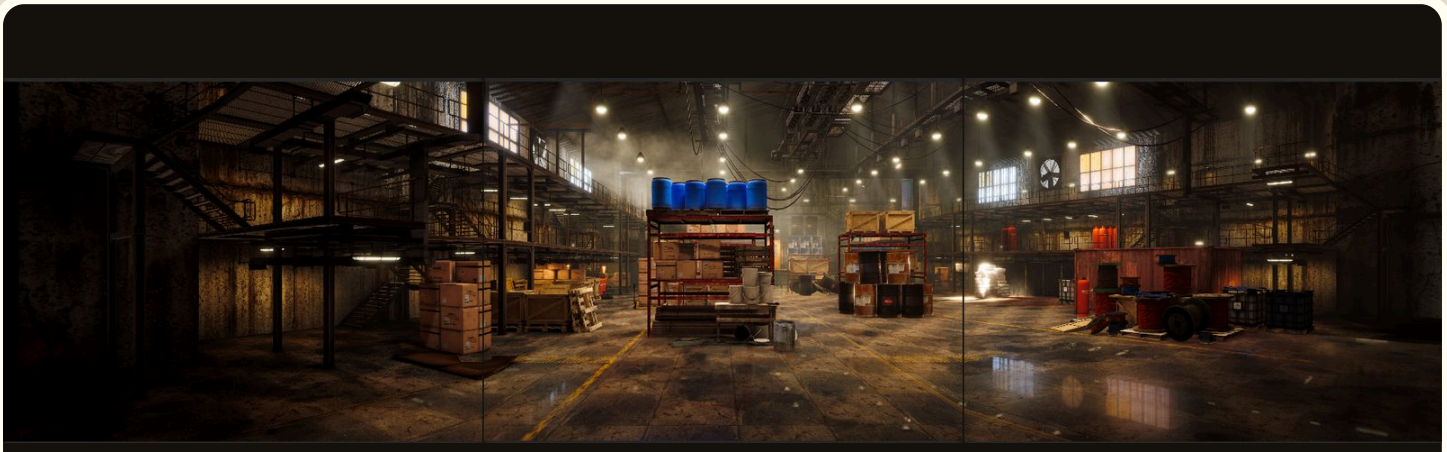
- B.Sc. Computer Engineering — Shahid Beheshti University (2016–2021)
- M.Sc. IT Management — University of Tehran, Markazi Branch (2021–2023)

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Industrial Environment in Unity

Industrial Warehouse • Unity • URP

Melika Shahghadami | Technical Artist | Environment Art + Volumetric Lighting



Main work — industrial warehouse environment built in Unity URP

About the Project

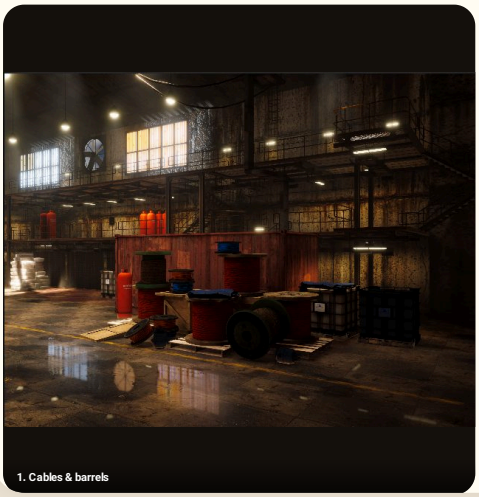
Industrial warehouse environment built in Unity using URP, focusing on PBR materials, volumetric lighting, environment art, and the design and layout of industrial props. The scene is optimized for real-time rendering and use in game projects.

Unity · URP · Environment · Lighting · PBR · Level Art

Technical Specs

URP	Volumetric	Game Ready
Render Pipeline	Volumetric light & fog	Optimized for games

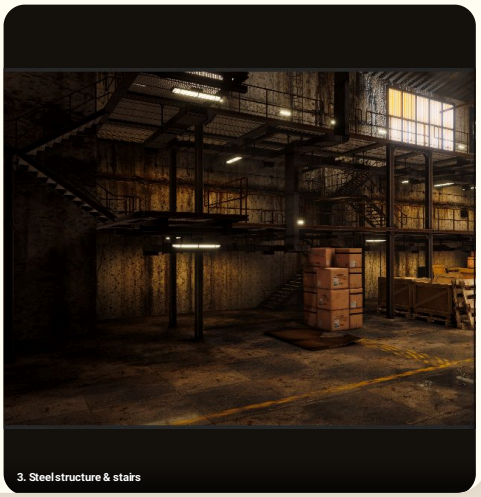
Environment details — 3 warehouse angles — Unity URP



1. Cables & barrels



2. Shelving & props



3. Steel structure & stairs

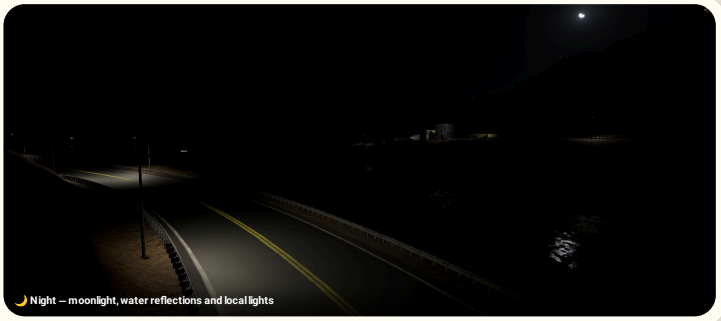
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Multiplayer VR Driving Simulator — Day/Night Cycle

Multiplayer VR Driving Simulator • Day / Night Cycle • Meta Quest 3

Melika Shahghadami | Technical Artist | Large-scale multiplayer environment with dynamic lighting


☀️ Day — mountain lake with natural light and ambient detail


🌙 Night — moonlight, water reflections and local lights

About the Multiplayer Project

Multiplayer VR driving simulator for Meta Quest 3 with a shared environment and day/night cycle. The environment includes a mountain road, tunnel, lake, vegetation and structures, with dynamic lighting and real-time optimization targeting 72+ FPS.

Multiplayer · VR 72+ FPS · Day/Night · Network Ready · Environment Art · Lighting · Unity · Quest 3

☀️ Day Views — Route Variety


Billboard


Aerial


Mountains

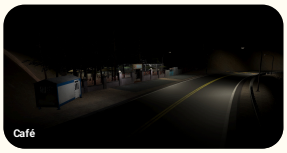

Forest


Dirt Road

🌙 Night Views — Lighting System


Tunnel


Speed Bump


Café


Lit Billboard


Building

🔄 Day/Night Comparison — One Location, Two Lightings


Day — Shoreline

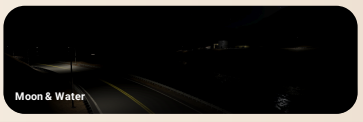

Night — Café


Day — Dirt Road


Day — Hill


Panorama


Main Road


Moon & Water

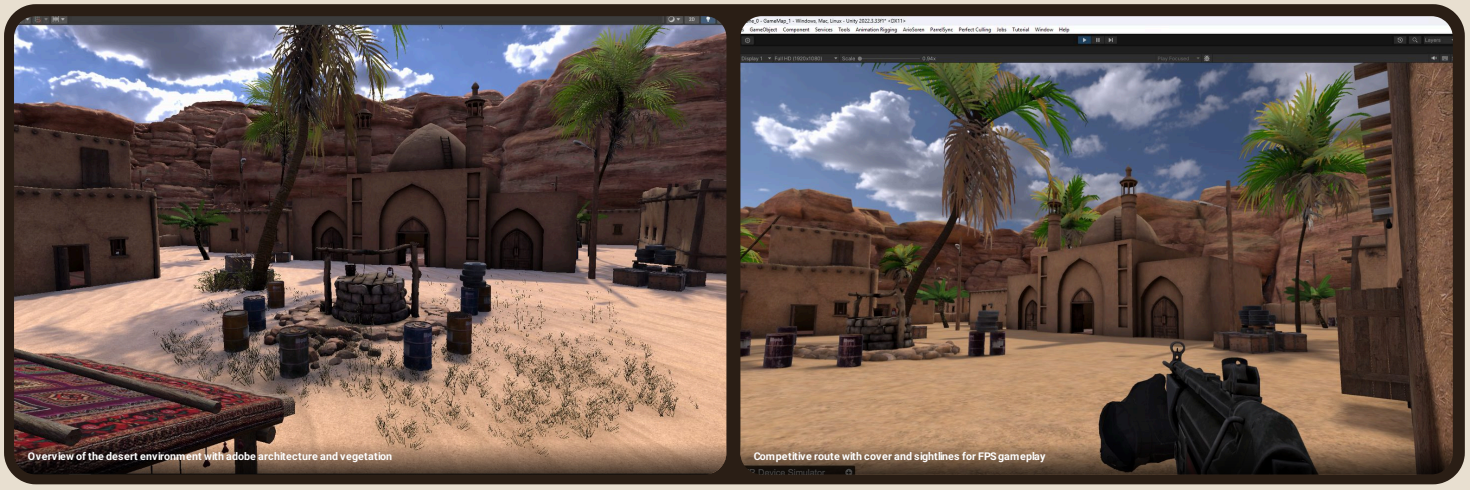

Night Tunnel

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Multiplayer Shooter Environment

Multiplayer Shooter Environment • Unity • FPS • Level Design

Melika Shahghadami | Technical Artist | FPS environment design focused on multiplayer gameplay



Overview of the desert environment with adobe architecture and vegetation

Competitive route with cover and sightlines for FPS gameplay

About the Project

First-person multiplayer shooter environment set in a desert with adobe architecture and palm groves. As technical artist, I was responsible for level layout, optimization for multiplayer, and route design with cover and sightlines for balanced gameplay. The environment is designed in Unity for multiplayer FPS, focusing on route readability, player flow and performance.

Multiplayer | FPS | Level Design | Unity | Environment Art | Optimization



Cover & Side Route View

Competitive View – Main Route



Night – Village

Night – Well & Lights

Interior – Traditional Room

Cinematic 3D Environment

Cinematic 3D Environment • Blender • Cycles • Atmospheric Lighting

Melika Shahghadami | Technical Artist | Main piece + lighting variant



Main work — full cinematic environment with windmill, abandoned hut, stone tower and lake — Blender Cycles

About the Project

Personal cinematic environment built in Blender combining a windmill, abandoned hut, stone tower and lake. The scene is created with Cycles, PBR materials, atmospheric lighting, volumetric fog and particle systems, and features water with reflections and displacement.

Blender | Cycles | PBR | Main | Lighting Variant | Cinematic

Lighting variant — lake and reeds — Blender / Cycles



Lake and reeds — overcast day, still water

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Wolf Character — Modeling & Fur

Wolf Character • Blender Particle Hair • Fur Grooming • PBR Texturing

Melika Shahghadami | Technical Artist | Modeling + Particle Hair / Fur Grooming





Final render — three-quarter view with studio lighting and fur detail

Final render — side view with PBR texturing and fur grooming

About the Project

Wolf character project focused on high/low poly modeling, retopology, UVs, PBR texturing and fur grooming in Blender. The fur is built with particle hair and grooming, and the model is prepared for both real-time and offline rendering.

Blender | Particle Hair | Fur Grooming | PBR | Retopology | Character Art | Organic Modeling

Technical Features

- Low/high poly modeling with optimized retopology
- UV and PBR texturing with fur detail
- Particle hair and fur grooming in Blender

Artistic Achievements

- Organic character modeling focused on anatomy and surface detail
- Particle hair and fur grooming for rendering
- Studio lighting and final presentation

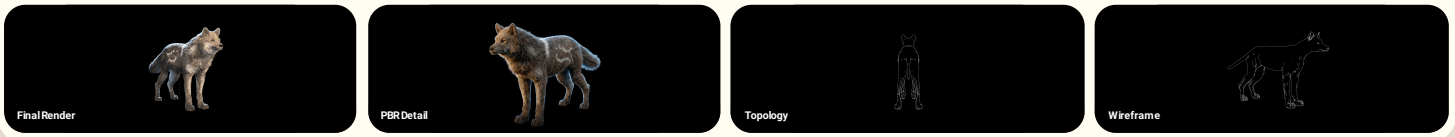
Wireframe & Topology



Wireframe — back view with optimized topology and clean edge flow

Wireframe — three-quarter view showing anatomical structure

Technical Details — Modeling, Texturing & Grooming



Final Render

PBR Detail

Topology

Wireframe

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Hard Surface Modeling

Hard Surface Modeling • Blender • PBR • Game Ready

Melika Shahghadami | Technical Artist | 4 Industrial Models





Energy Battery Pack



FUEL Barrel

12K–20K Tris | Game Ready | PBR | Hard Surface



1. Energy Battery Pack



2. Fuel Barrel



3. Industrial Toolbox



4. Industrial Robotic Arm

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